19.3, 19.4 The Divergence Theorem

→ Do problems 1-3.

If $\vec{F}$ is a vector field, the divergence of $\vec{F}$ at a point is

(strength of field leaving point) − (strength of field entering).

To compute it, we use the formula

$$\text{divergence} = \nabla \cdot \vec{F}.$$

→ Do problem 4.

If you add up the divergence at each point in a solid W (in $\mathbb{R}^3$), you'll get the flux through the boundary of W. This is:

Divergence Theorem If W is a solid in $\mathbb{R}^3$ whose boundary S is a piecewise smooth surface, and if $\vec{F}$ is a smooth vector field, then

$$\int_S \vec{F} \cdot d\vec{A} = \int_W \nabla \cdot \vec{F} \, dV$$

where S is given the outward orientation.

→ Do problems 5-7.